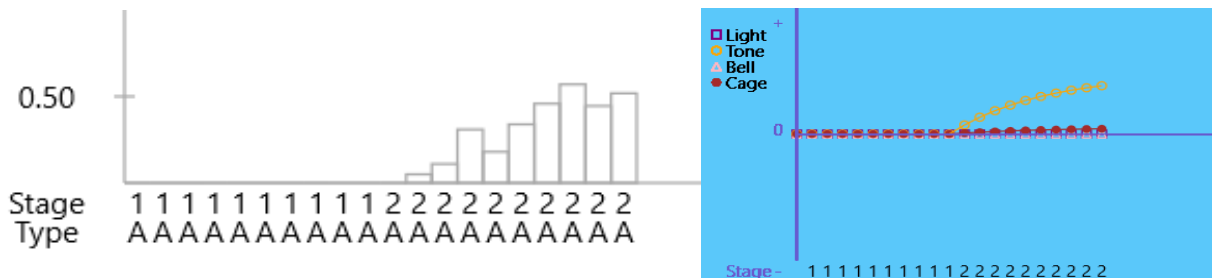


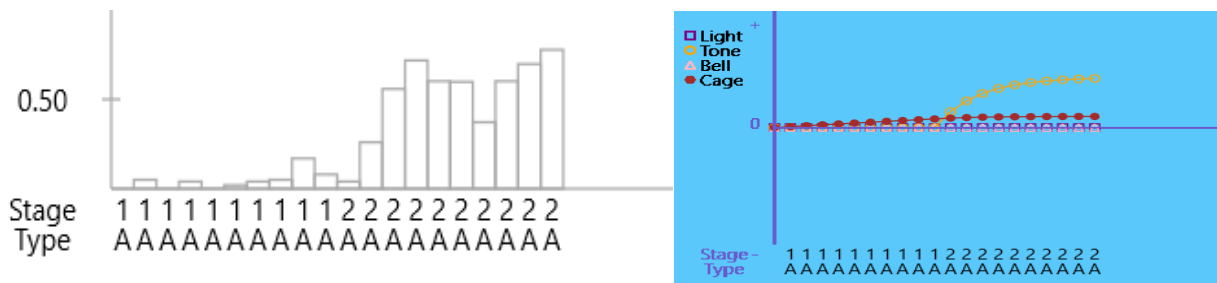
Sniffy: Project's #4, # 5, #6, and #7

Sniffy: Project #4

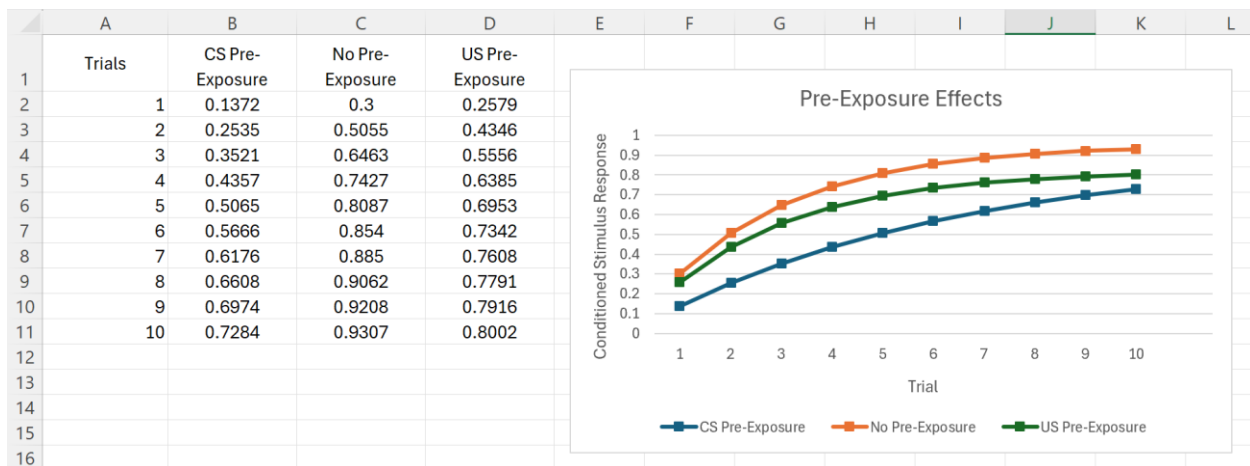
Exercise #20: “Movement Ratio” and “CS Response Strength” graphs



Exercise #21: “Movement Ratio” and “CS Response Strength” graphs



Excel graph that compares the effects of “US” and “CS” pre-exposure



Interpretation of these results:

- The resulting data from exercises 20 and 21 are quite similar. Both show a gradual increase in the “Movement Ratio” graph as well as the “CS Response Strength” graph. However, looking specifically at the “CR Response Strength” graph for exercise 21, we notice that the slope/increase is slightly more pronounced than it is in exercise 20. They both reach around the same maximum value, but it does so quicker in the “CS Response” graph for exercise 21 and then kind of plateaus (increased conditioning efficiency). Additionally, looking at the “Movement Ratio” graph for exercise 21, it can be seen that the increase in observable fear/freezing behaviors is noticeably less gradual (the number of freezing behaviors increases quicker than in exercise 21) (increased learning rate). Together these results, as shown in the excel graph, show that pre-exposure to either a CS or a US will lead to a decreased learning rate and decreased conditioning efficiency. Furthermore, we can see that while learning rate/efficiency were lowered in comparison to the control when pre-exposed to EITHER the CS or US, Sniffy learned quicker/better when pre-exposed to the US rather than the CS (CS has the worst results, learning process was most impaired by CS pre-exposure).

Q.01: “How does pre-exposure to stimuli relate to learning in the real-world?”

- A: Pre-exposure to a stimulus often makes it harder to later learn an association with that stimulus because the brain has already filed it away as irrelevant, a phenomenon known as latent inhibition. In real life, this means that familiar signals we’ve repeatedly encountered without consequence are more likely to be ignored when they eventually gain importance—for example, a brand logo seen countless times without reward is less effective in advertising, or a teacher’s overused signal loses its power to command attention. Essentially, learning is shaped not just by what is paired together, but by the prior history of exposure, with organisms prioritizing novelty and surprise over what has long seemed meaningless

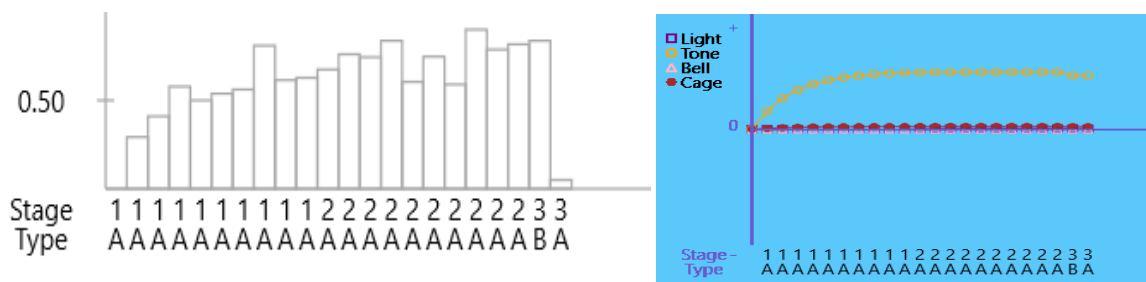
Q.02: “What, if any, differences occur with respect to pre-exposure to a CS or US?”

- A: The rate and efficiency of learning/conditioning are slightly better when pre-exposed to the US (rather than pre-exposure to the CS). Its hard to notice, but in exercise 21 (pre-exposure to the US) we can see a barely noticeable increase in the “CS Response Strength” to the “cage” specifically (not just the “tone” as seen with

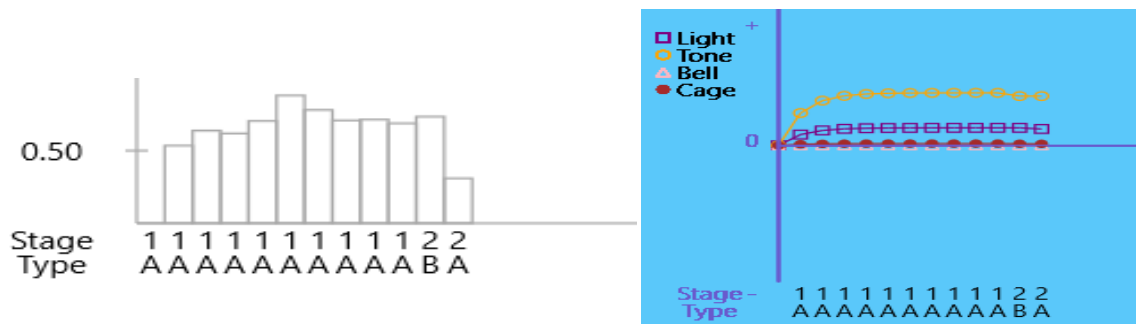
the exercise 20 graph). This is because the rat in exercise 21 did not have a “tone” to associate the shock with during pre-exposure, so it associated the shock with the “cage.” The rat in exercise 20 did not even have a shock during pre-exposure, so it did not associate the cage with it (the US). Also, we can generally see a larger negative effect on learning when pre-conditioned to CS rather than a US.

Sniffy: Project # 5

Exercise #07: “Movement Ratio” and “CS Response Strength” graphs



Exercise #08: “Movement Ratio” and “CS Response Strength” graphs

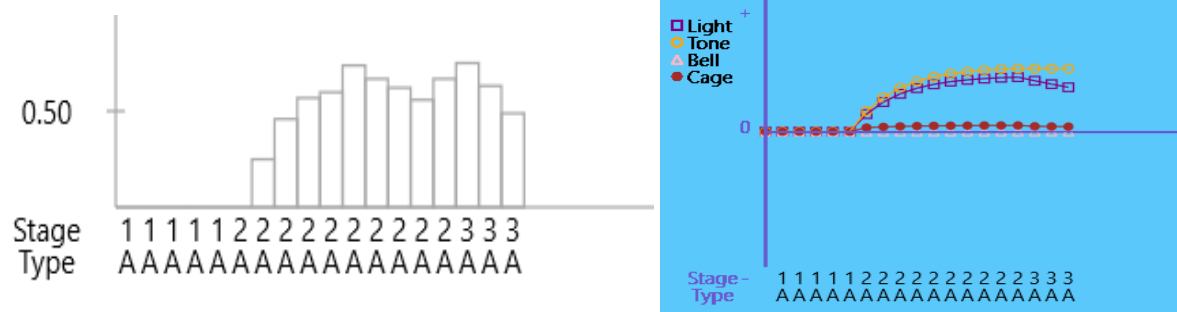


Interpretation of these results: The data from the “Movement Ratio” and “CS Response Strength” graphs shows us two things. The first is that pairing one specific type of CS (“tone”) with a US, before presenting both types of CS with the US, leads to a conditioned response for only that CS (increase in “CS Response Strength” only observed in “tone”). This is evidence for the process of “blocking” which occurs in “compound conditioning.” Secondly, the data shows that an increase in salience for one specific CS during pairing leads to a greater conditioned response strength than the other one when both are presented later with no US. This is evidence that “overshadowing” really does occur in the

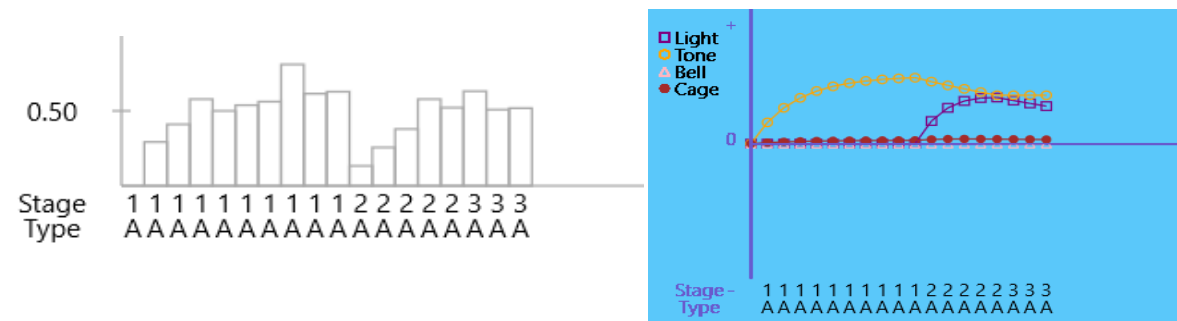
real world. Together these tell us that the order of stimuli presented, as well as their salience, play important roles in the process of conditioning and learning.

Sniffy: Project # 6

Exercise #12: “Movement Ratio” and “CS Response Strength” graphs



Exercise #13: “Movement Ratio” and “CS Response Strength” graphs



Interpretation of these results: Comparing the results from the graphs of these two exercises together, there is really only two primary differences seen among the data sets. For exercise 21, there is “Movement Ratio” and “CS Response Strength” data being generated during “Stage 1,” whereas none is generated during “Stage 1” for exercise 20. Secondly, we see a simultaneous increase in “CS Response Strength” for “tone” and “light” at the beginning of “stage 2” for exercise 20. In exercise 21, a staggered increase is seen with the “tone” increasing before “light” at the very beginning of “stage 1.” Both of these outcomes are due to the fact that the two CS types were associated together either before or after the CR was conditioned to the CS-2. Similar to project #5’s interpretation of the results, this data really highlights the importance of sequential order for the CS’s

Interpretation of these results: The results from both these exercises indicates that learning/conditioning occurs slower. Looking at stage 1, we see that the “Movement Ratio” increases for trial A but not trial B. Then, looking at the “CS Response Strength” in stage 2 we see the inhibitory conditioning slowed the learning of a fear response to the shock